SECTION 3 · SUBSECTION 3.7
DOCUMENT GENERATION

3.7.1 INTRODUCTION.

The document generator is a pure function over state. It assembles the onboarding document section by section, produces a compact context document for a downstream AI assistant, serialises the complexity report, and writes a per-file explanation pair in Markdown and JSON. The one idea the reader must leave with is that every section is independently guarded: one failed section is omitted with a warning, and the remaining document is still returned (figure 3.7-1).

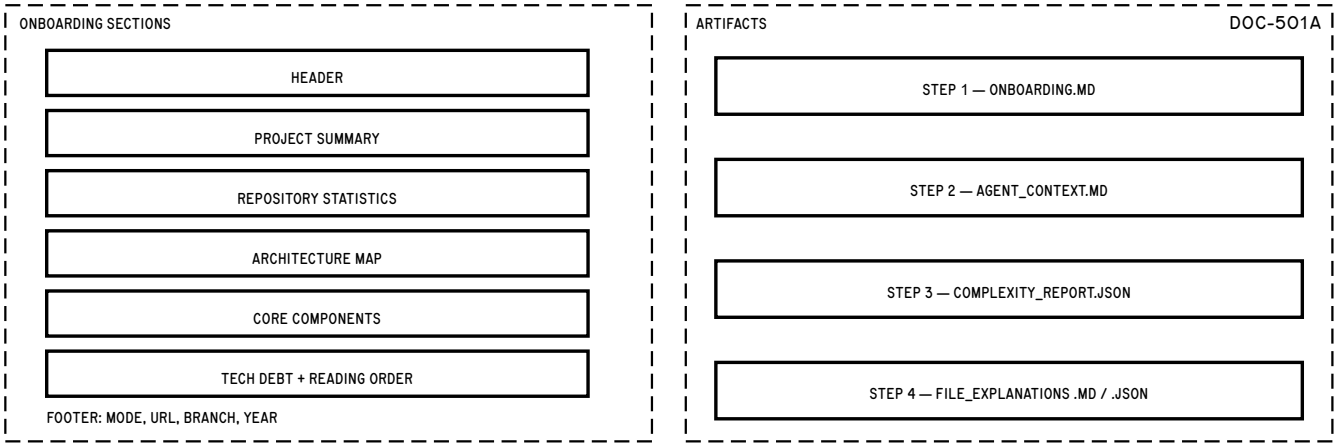



Figure 3.7-1. Document Assembly and Output Artifacts

3.7.2 FUNCTIONAL DESCRIPTION.

*run* iterates the onboarding builders in a fixed order, appending each returned string. It then calls *generate_agent_context_doc*, which composes ten sections aimed at an AI coding assistant: overview, entry points, core components, relationships, high-risk areas, cycles, domain objects, change impact, safe starting points and AI explanations. The complexity report is serialised from the score objects. The per-file explanation document and JSON are produced by separate builders. Each of the four major outputs is wrapped in its own try block.

3.7.2.1 Degradation Rule.

Section builders that fail are skipped, not replaced by a placeholder, so a document can be missing a heading without any visible marker in the body. A failure of the whole agent-context builder substitutes a one-line error document. The API always returns whatever fields were produced; the CLI writes only the files that exist.

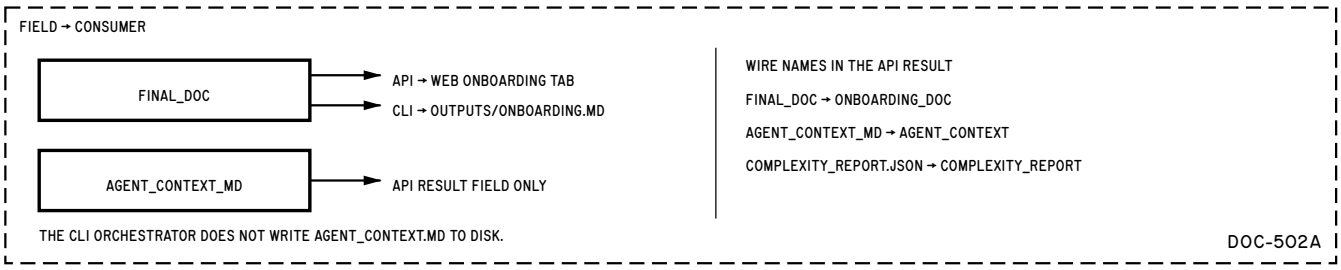



Figure 3.7-2. Output Field Consumers

3.7.3 PERFORMANCE AND DESIGN DATA.

Table 3.7-1. Generated Documents

| Artifact | Content | Written by |
|---|---|---|
| onboarding.md | Eight narrative sections | CLI and API result |
| agent_context_md | Ten assistant-oriented sections | API result field only |
| complexity_report.json | Per-file scores and reasons | CLI and API |
| file_explanations.md | One section per explained file | CLI save helper |
| file_explanations.json | Path to explanation map | CLI save helper |
| symbol_tables.json | All parsed symbols | Orchestrator save |
| dependency_graph.json | Nodes, edges, cycles, order | Orchestrator save |

**CAUTION**

The "Important Domain Objects" section reads *cls.methods*, but the class record exposes *method_names*. Every class therefore reports zero methods. See 8.1-1. Separately, the CLI path never persists *agent_context_md*; only the API returns it.

```
DOCUMENT CHECK                    REVIEWER
1 OPEN - onboarding.md
2 COUNT - major headings present
  √ RISK - table matches scores
3 CHECK - reading order not empty
4 VERIFY - footer names branch and mode
```